

CSE331 - Automata and Computability

Quiz 2

Section 16 (Set A)

Full name:

ID:

Total Marks: 20

1. (3 + 3 + 4 marks) Write the regular expressions based on the following:

L_1 = Starts and ends with 1

L_2 = Length is at least 4

L_3 = Contains 10 as a substring

- a. Regular expression for L_1 , L_2 and L_3

- b. Regular expression for $\overline{L_1}$, $\overline{L_2}$, and $\overline{L_3}$

- c. Regular expression for $\overline{L_1 \cap L_3}$ and $\overline{L_1 \cup L_2}$

Answer guidelines:

- a. L_1 : an edge case of length 1 is to be considered.

L_2 : Done in class

L_3 : Done in class

- b. L_1' : can start and end with 0, different symbols, and edge cases of length 0 and 1

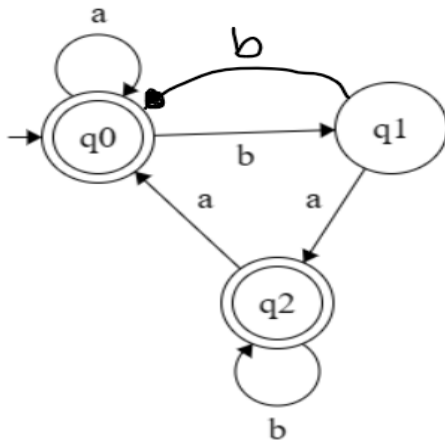
L_2' : length at most 3

L_3' : here 0 cannot appear after 1, so answer is 0^*1^*

- c. $\overline{L_1 \cap L_3} = \overline{L_1} \cup \overline{L_3}$

$\overline{L_1 \cup L_2} = \overline{L_1} \cap \overline{L_2}$ which means that strings of length at most 3 which don't start and end with 1

2. (10 marks) Convert the following FA into RE in the order q2, q0, q1:



Preprocessing step.

Eliminating q2 transitions: $\{q1\} \times \{q0, f\}$

Eliminating q0 transitions: $\{s, q1\} \times \{q1, f\}$

Eliminating q1 transitions: $\{s\}, \{f\}$

You need to show state diagram after each elimination. At the end there must be only one starting state and only one accepting state.

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Quiz 2

Section 16 (Set B)

Full name:

ID:

Total Marks: 20

1. (3 + 3 + 4 marks) Write the regular expressions based on the following:

L_1 = Starts and ends with 0

L_2 = Length is at most 5

L_3 = Contains 01 as a substring

a. Regular expression for $\overline{L_1}$, $\overline{L_2}$ and $\overline{L_3}$

b. Regular expression for $\overline{L_1}$, $\overline{L_2}$, and $\overline{L_3}$

c. Regular expression for $\overline{L_1 \cap L_3}$ and $\overline{L_1 \cup L_2}$

Answer guidelines:

a. L_1 : an edge case of length 1 is to be considered.

L_2 : Done in class

L_3 : Done in class

b. L_1' : can start and end with 1, different symbols, and edge cases of length 0 and 1

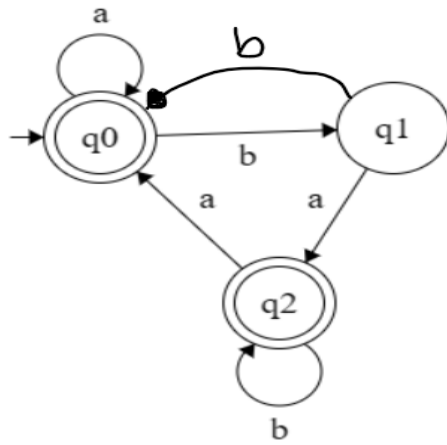
L_2' : length at least 6

L_3' : here 1 cannot appear after 0, so answer is 1^*0^*

c. $\overline{L_1 \cap L_3} = \overline{L_1} \cup \overline{L_3}$

$\overline{L_1 \cup L_2} = \overline{L_1} \cap \overline{L_2}$ which means that strings of length at least 6 which don't start and end with 0

2. (10 marks) Convert the following FA into RE in the order q1, q0, q2:



Preprocessing step.

Eliminating q1 transitions: $\{q0\} \times \{q0, q2\}$

Eliminating q0 transitions: $\{s, q2\} \times \{f, q2\}$

Eliminating q2 transitions: $\{s\}, \{f\}$

You need to show the state diagram after each elimination. At the end there must be only one starting state and only one accepting state.